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With the Gigahertz wars underway, the need for faster and hard drives was expected. We test the best hard drives available in the Indian market, to see which of them compliment your configuration, as well as your pocket

Every year, most computer users spend money on one particular component that is most fragile, and at the highest risk. It puts companies, and home-users alike, on pins and needles when it starts acting up, which just goes to show how important it is. It's the hard drive—a versatile digital storage medium.

Today, if you conduct a survey, you will notice that the minimum capacity of the hard drive that your vendor is likely to put in your system is 40 GB, unless you specify a different capacity. Does any one require such a mammoth digital storage space? Well, when you think about it, we don't even require the latest, and fastest Pentium 4 processors. It may not be the need of the hour, but it is always advisable to make your PC as future proof as possible, considering that the default installation of Windows XP needs about 1.2 GB—only for the operating system—then there are applications that increase in size with every new version. The latest games require at least 1 GB of space, the vast MP3 collection that you are going to collect over time, and last but not least movies—all these space mongering collectibles are sure to fill up a 250 GB hard disk in no time. Although, as of now, a 250 GB hard drive seems too large to be even considered, it will soon become a standard accepted level. Remember, what people said about the 40 GB hard drives when they were launched, while 4 GB was still the norm?

So why are we doing a HDD test? A gamer would like his hard drive to perform well enough to load 3D games faster. A multimedia professional needs to see the load time of his heavy graphic software dwindling, while a normal user just wants to get movies, documents, and songs copied, and ripped faster than ever before. With hyper-threading already on desktop PCs, multi-tasking has acquired a new dimension, and the hard drive subsystem needs to complement, and keep pace with latest components. Since SCSI is not only complicated, but also costly for desktop systems, one has to settle for IDE, or the latest Serial ATA drives.

However, in no way should these

drives be underestimated. An IDE 133 drive can load applications, transfer data at high speeds. The advantage of higher capacities, and hard drives is that you get optimum for storage space. The per-rupee value of every MB of the data that you store on a hard drive, has come down considerably. Ironically, you can get a higher capacity hard drive for the same amount spent on your existing hard drive. To calculate the capacity of the hard drive and make a note of the amount spent on it, you can come up with a GB to MB ratio that will help you buy a hard drive that is within your budget.

This article will provide you with the insight needed to arrive at a purchase decision when a new hard drive becomes the need of the hour. We tested various brands of hard drives, with capacities ranging from the current entry-level to the humongous ones. The hard drives that we have tested are the older IDE drives, and the newer Serial ATA drives, and also provide you with buying tips to enable you to make an informed buying decision and enable you to choose a drive that meets your needs for a long time to come.

Test process

The IDE and Serial ATA (SATA) drives were tested on a test PC that comprised a Pentium 4 3.2 GHz, on an Intel i875 (Everblood) motherboard, with 512 MB Dual Channel 400 MHz DDR SDRAM. The video card used was a GeForce 4 Ti 4200 with 128 MB of DDR memory. The IDE drives were connected to the port 0 SATA channel. All the IDE drives were connected to the primary IDE channel, while the SATA drives were connected to the Primary IDE channel. The necessary drivers and benchmarking utilities were loaded on the drive after installing Windows XP. The latest drivers for the main board and IDE channel were installed. The results of the benchmarks used to evaluate the performance of the hard disk are given below.

HDTach 2.6

In this test, sequential read and write speed, read burst speed, average access time, and CPU utilisation are noted down. The results of the first in the battery of tests